

Offline Repository

To verify the server operating system and version, the following command was executed:

```
cat /etc/os-release
```

This confirms that the server is running **CentOS Stream 9**, a modern, rolling-release Linux distribution

Operating suitable for enterprise environments and development of production-ready systems.

System Identity and User Validation

To verify system configuration, user privileges, and operating system details, the following commands were executed:

```
id ronaldo  
hostnamectl
```

User Verification

Observed Output:

```
uid=1000 (ronaldo)  
  
gid=1000 (ronaldo)  
  
groups=1000 (ronaldo) ,10 (wheel)
```

Interpretation:

The user **ronaldo** exists and is properly configured

Membership in the **wheel group** confirms that the user has **sudo (administrative) privileges**

This allows controlled privilege escalation for system administration tasks

System and Host Information

Observed Output from hostnamectl:

Hostname: lamp-ansible

Operating System: CentOS Stream 9

Kernel Version: Linux 5.14.x

Architecture: x86_64

Virtualization: Oracle VirtualBox

Overview The server was initially installed with a graphical user interface (GUI). However, the system has now been transitioned to **multi-user.target (CLI-based mode)**

```
[ronaldo@lamp-ansible ~]$ sudo systemctl set-default multi-user.target
```

Previous Operating Mode:

graphical.target (Runlevel 5 – GUI mode)

```
[ronaldo@lamp-ansible ~]$ sudo -i
[sudo] password for ronaldo:
[root@lamp-ansible ~]#
[root@lamp-ansible ~]# su - ronaldo
[ronaldo@lamp-ansible ~]$
[ronaldo@lamp-ansible ~]$
[ronaldo@lamp-ansible ~]$
[ronaldo@lamp-ansible ~]$ sudo systemctl set-default multi-user.target
```

Transition to Production Mode:

The system was reconfigured to operate in a server-optimized, non-GUI environment using the following commands:

sudo systemctl set-default multi-user.target

```
[ronaldo@lamp-ansible ~]$ sudo systemctl set-default multi-user.target
[sudo] password for ronaldo:
Removed "/etc/systemd/system/default.target".
Created symlink /etc/systemd/system/default.target → /usr/lib/systemd/system/multi-user.target.
[ronaldo@lamp-ansible ~]$
```

Sudo reboot

```
[ronaldo@lamp-ansible ~]$ sudo systemctl set-default multi-user.target
[sudo] password for ronaldo:
Removed "/etc/systemd/system/default.target".
Created symlink /etc/systemd/system/default.target → /usr/lib/systemd/system/multi-user.target.
[ronaldo@lamp-ansible ~]$ sudo reboot
```

This ensures that the server boots into **multi-user.target (Runlevel 3 – CLI mode)**, which is the standard for production environments due to improved performance, reduced resource utilization, and enhanced security.

Verification of Configuration:

After rebooting, the system state was verified using:

```
Sudo systemctl get-default
runlevel
```

Expected Output:

```
CentOS Stream 9
Kernel 5.14.0-661.el9.x86_64 on an x86_64

Activate the web console with: systemctl enable --now cockpit.socket

lamp-ansible login: ronaldo
Password:
Last login: Wed Apr 29 23:42:12 on pts/0
[ronaldo@lamp-ansible ~]$ ls
Desktop  dh1  dh2  dh3  dh4  dh5  Documents  Downloads  Music  Pictures  Public  Templates  Videos
[ronaldo@lamp-ansible ~]$
[ronaldo@lamp-ansible ~]$ system get-default
bash: system: command not found...
[ronaldo@lamp-ansible ~]$ sudo system get-default
[sudo] password for ronaldo:
sudo: system: command not found
[ronaldo@lamp-ansible ~]$ sudo systemctl get-default
multi-user.target
[ronaldo@lamp-ansible ~]$ runlevel
N 3
[ronaldo@lamp-ansible ~]$ _
```

multi-user.target

This confirms that the system is now running in CLI mode and is properly aligned with production-ready server deployment standard

Perform system upgrade `sudo dnf upgrade -y`

Before installing the LAMP stack, I executed `sudo dnf upgrade -y` to ensure the server had the latest security patches. The system returned a message indicating it was already up to date, confirming a stable and secure baseline for the deployment."

```
[ronaldo@lamp-ansible ~]$ sudo dnf upgrade -y
[sudo] password for ronaldo:
Last metadata expiration check: 0:53:52 ago on Thu 30 Apr 2026 06:33:27 PM SAST.
Dependencies resolved.
Nothing to do.
Complete!
[ronaldo@lamp-ansible ~]$
```

Security Validation

SELinux Status Check:

getenforce

Expected Output:

Enforcing

```
[ronaldo@lamp-ansible ~]$ getenforce
Enforcing
[ronaldo@lamp-ansible ~]$
```

Explanation:

SELinux enforces security policies

Protects the web server from unauthorized access

Firewall Configuration Check:

firewall-cmd --list-all

is active

Ensure:

Required services (HTTP/HTTPS) will be allowed after deployment

Resource Availability and System Readiness Check

Disk Space Verification

Command:

df -h

Requirement: Sufficient disk space must be available to support:

Apache log files

Database storage (MariaDB)

WordPress application files

Memory Usage Check

Command:

free -m

Importance:

Ensures the system can handle web server and database workloads

Prevents performance degradation under load

Service Management Readiness

The system uses **systemd** to manage critical services such as Apache and MariaDB.

Verification Command:

sudo systemctl list-units --type=service

```
Complete!
[ronaldo@lamp-ansible ~]$ sudo systemctl list-units --type=service
[sudo] password for ronaldo:
UNIT                                LOAD    ACTIVE SUB    DESCRIPTION
alsa-state.service                 loaded active running Manage Sound Card St
atd.service                         loaded active running Deferred execution s
auditd.service                     loaded active running Security Auditing Se
avahi-daemon.service               loaded active running Avahi mDNS/DNS-SD St
chronyd.service                    loaded active running NTP client/server
crond.service                       loaded active running Command Scheduler
cups.service                       loaded active running CUPS Scheduler
dbus-broker.service                loaded active running D-Bus System Message
dracut-shutdown.service             loaded active exited Restore /run/initram
firewalld.service                  loaded active running firewalld - dynamic
getty@tty1.service                 loaded active running Getty on tty1
irqbalance.service                loaded active running irqbalance daemon
kdump.service                      loaded active exited Crash recovery kerne
kmod-static-nodes.service           loaded active exited Create List of Stati
libstoragemgmt.service              loaded active running libstoragemgmt plug-
lvm2-monitor.service                loaded active exited Monitoring of LVM2 m
mcelog.service                     loaded active running Machine Check Except
ModemManager.service                loaded active running Modem Manager
NetworkManager-wait-online.service loaded active exited Network Manager Wait
NetworkManager.service              loaded active running Network Manager
```

Observed State: Most services are:

Loaded

Active

Running

Workstation Connectivity & Remote Management.

Windows PowerShell SSH client. This allows for seamless command-line management of the CentOS9 server directly from the workstation

PHASE1: OFFLINE REPOSITORY (MANUAL CONFIGURATION)

Objective:

To establish a local software source using the CentOS Stream 9 DVD ISO, enabling the installation of the LAMP stack (Apache, MariaDB, PHP) without an external internet connection

A. What is an Offline Repository?

An offline repository is a **local package source** that replaces internet-based package downloads.

Used in:

Secure environments (banks, government systems)

Air-gapped servers

Enterprise Linux infrastructure

Why It Is Important

No internet dependency

Faster package installation

Controlled software versions

Stable enterprise deployments

2.B.CentOS 9 ISO Mounting and Troubleshooting Report

Mount ISO

```
mount -o loop CentOS-Stream-9.iso /mnt
```

Understand ISO Structure

/mnt

└─ BaseOS → Core OS packages

└─ AppStream → Application packages

1. Introduction

This document explains the process of downloading, verifying, and mounting a CentOS Stream 9 ISO image on a Linux system. It also outlines the challenges

encountered during the process, including failed downloads, network troubleshooting, and mounting errors, along with the solutions applied.

2. Initial Attempt to Mount ISO

The process began with an attempt to mount an ISO file:

```
sudo mount -o loop /home/ronaldo/myfile.iso /mnt/iso
```

Issues Encountered

File not found errors

“Cannot find fstab” message

Failure to set up a loop device

Analysis

These errors indicated that:

The ISO file either did not exist or was invalid

The mount command syntax or file path was incorrect

3. Verification of ISO File

To confirm file availability:

```
ls -lh /home/ronaldo
```

Observation:

Several files existed but were very small (4KB–11KB)

These were identified as **incomplete or corrupted downloads**, not valid ISO images

4. Network Troubleshooting .

Before retrying downloads, network connectivity was verified to rule out issues with DNS, routing, or internet access.

DNS and Connectivity Checks

Test 1: Ping a public DNS server

```
ping 8.8.4.4
```

Test 2: Ping a domain name

```
ping google.com
```

Test 3. Check interface.

```
ip a
```

Purpose

Confirm basic internet connectivity (IP reachability)

Confirm DNS resolution is working (domain name translation)

Confirm the interface

DNS Configuration Adjustment

To further troubleshoot potential DNS issues, the DNS server was changed from:

8.8.4.4 (Google DNS)

to:

1.1.1.1 (Cloudflare DNS)

Reason

To eliminate DNS-related causes of failed **wget** downloads

To improve reliability of domain resolution

Outcome

Network and DNS functionality confirmed to be working correctly

Download failures were not caused by connectivity issues

5. ISO Download Process Using Wget

Initial **wget** attempts failed due to:

404 errors

Incorrect URLs

Incomplete downloads

Final Solution

The ISO was located and copied from a verified mirror:

```
https://centos-stream.mirror.liquidtelecom.com/9-  
stream/BaseOS/x86_64/iso/CentOS-Stream-9-latest-x86_64-  
dvd1.iso
```

Downloaded using:

```
wget -c https://centos-  
stream.mirror.liquidtelecom.com/9-  
stream/BaseOS/x86\_64/iso/CentOS-Stream-9-latest-  
x86\_64-dvd1.iso
```

```
[ronaldo@lamp-ansible ~]$ wget -c https://centos-stream.mirror.liquidtelecom.com/9-stream/BaseOS/x86_64/iso/CentOS-Stream-9-latest-x86_64-dvd1.iso  
--2026-05-01 18:37:08-- https://centos-stream.mirror.liquidtelecom.com/9-stream/BaseOS/x86_64/iso/CentOS-Stream-9-latest-x86_64-dvd1.iso  
Resolving centos-stream.mirror.liquidtelecom.com (centos-stream.mirror.liquidtelecom.com)... 197.155.77.1, 2c0f:fe40:8001:10::1  
Connecting to centos-stream.mirror.liquidtelecom.com (centos-stream.mirror.liquidtelecom.com)|197.155.77.1|:443... connected.  
HTTP request sent, awaiting response... 200 OK  
Length: 15378677760 (14G) [application/octet-stream]  
Saving to: 'CentOS-Stream-9-latest-x86_64-dvd1.iso'  
  
CentOS-Stream-9-latest-x86_64-dvd 100%[=====] 14.32G 3.48MB/s in 71m 49s  
2026-05-01 19:48:58 (3.40 MB/s) - 'CentOS-Stream-9-latest-x86_64-dvd1.iso' saved [15378677760/15378677760]
```

Result

Successful download to **/home/ronaldo**

```
[ronaldo@lamp-ansible ~]$ ls -lh  
total 15G  
-rw-r--r--. 1 ronaldo ronaldo 15G Apr 27 06:18 CentOS-Stream-9-latest-x86_64-dvd1.iso
```

File size confirmed (~15GB), indicating a valid ISO

6. Mount Point Preparation

```
sudo mkdir -p /mnt/iso
```

7. ISO Mounting

```
sudo mount -o loop /home/ronaldo/CentOS-Stream-9-latest-x86_64-dvd1.iso /mnt/iso
```

System Response

```
[ronaldo@lamp-ansible ~]$ sudo mkdir -p /mnt/iso
[ronaldo@lamp-ansible ~]$ sudo mount -o loop /home/ronaldo/CentOS-Stream-9-latest-x86_64-dvd1.iso /mnt/iso
mount: /mnt/iso: WARNING: source write-protected, mounted read-only.
```

Warning: source is write-protected, mounted read-only

This is expected behavior for ISO files

8. Verification of Mount

```
ls /mnt/iso
```

```
[ronaldo@lamp-ansible ~]$ sudo mount -o loop /home/ronaldo/CentOS-Stream-9-latest-x86_64-dvd1.iso /mnt/iso
mount: /mnt/iso: WARNING: source write-protected, mounted read-only.
[ronaldo@lamp-ansible ~]$ ls /mnt/iso
AppStream BaseOS EFI EULA extra_files.json images isolinux LICENSE media.repo
```

Output

BaseOS

AppStream

EFI

repodata

Conclusion

```
[ronaldo@lamp-ansible ~]$ ls -lh /mnt/iso
total 30K
drwxr-xr-x. 1 root root 2.0K Apr 27 06:13 AppStream
drwxr-xr-x. 1 root root 2.0K Apr 27 06:13 BaseOS
drwxrwxr-x. 1 root root 2.0K Apr 27 06:02 EFI
-r--r--r--. 1 root root 299 Apr 27 06:13 EULA
-r--r--r--. 1 root root 745 Apr 27 06:13 extra_files.json
drwxrwxr-x. 1 root root 2.0K Apr 27 06:02 images
drwxrwxr-x. 1 root root 2.0K Apr 27 06:02 isolinux
-rw-r--r--. 1 root root 18K Apr 7 23:09 LICENSE
-r--r--r--. 1 root root 88 Apr 27 06:13 media.repo
[ronaldo@lamp-ansible ~]$
```

The ISO was successfully mounted and is ready for use.

To make mount permanent:

(Auto-mount on Boot)

ISO file to be mounted every time to restart system.

ISO in the `/etc/fstab` file:

Opening the file: `sudo vi /etc/fstab`

Adding:

`/home/ronaldo/CentOS-Stream-9-latest-x86_64-dvd1.iso /mnt/iso iso9660`

`loop,ro 0 0`

Save and exit.

To verify: `cat /etc/fstab`

```
[ronaldo@lamp-ansible ~]$ sudo vi /etc/fstab
[sudo] password for ronaldo:
[ronaldo@lamp-ansible ~]$ cat /etc/fstab

#
# /etc/fstab
# Created by anaconda on Thu Feb 19 19:52:43 2026
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
/dev/mapper/cs-root    /                    xfs     defaults        0 0
UUID=8562a074-d5df-4e5e-8376-dac90ecd1869 /boot               xfs     defaults        0 0
/dev/mapper/cs-swap    none                swap    defaults        0 0

/home/ronaldo/CentOS-Stream-9-latest-x86_64-dvd1.iso /mnt/iso iso9660 loop,ro 0 0
[ronaldo@lamp-ansible ~]$
```

Mount is now permanent.

9. Key Challenges Faced

1. Invalid ISO Files

Presence of small, incomplete files misidentified as ISO images

2. Download Failures

Incorrect URLs and repeated `wget` failures

3. Network Uncertainty

Required DNS and connectivity testing to rule out infrastructure issues

4. Mount Errors

Loop device errors caused by invalid file references

Misinterpretation of `fstab` error messages

10. Resolution Summary

The issue was resolved through the following steps:

1. Verified network connectivity using ping tests
2. Adjusted DNS configuration to ensure proper resolution
3. Identified and removed invalid ISO files
4. Downloaded a valid ISO from a trusted mirror
5. Verified file size and integrity
6. Created a mount directory
7. Successfully mounted the ISO using a loop device
8. Verified mount by listing contents

11. Conclusion

This exercise demonstrated:

Effective troubleshooting of network and DNS issues

Proper validation of downloaded system images

Correct usage of loop devices in Linux

Practical system administration skills in a CentOS environment

The process highlights the importance of **systematic troubleshooting**, ensuring that both network and file integrity issues are resolved before performing system-level operations such as mounting ISO images.

STEP 3 — Create Local Repo Directory

Confirm iso file is still mounted;

findmnt /mnt/iso

```
[ronaldo@lamp-ansible ~]$ findmnt /mnt/iso
TARGET SOURCE FSTYPE OPTIONS
/mnt/iso /dev/loop0 iso9660 ro,relatime,nojoliet,check=s,map=n,blocksize=2048
[ronaldo@lamp-ansible ~]$
```

Output showing:

SOURCE → **your .iso file**

TARGET → **/mnt/iso**

mkdir -p /localrepo/BaseOS

mkdir -p /localrepo/AppStream

To verify:ls -l /localrepo

```
[ronaldo@lamp-ansible ~]$ sudo mkdir -p /localrepo/BaseOS
[ronaldo@lamp-ansible ~]$ sudo mkdir -p /localrepo/AppStream
[ronaldo@lamp-ansible ~]$ ls -l /localrepo
total 0
drwxr-xr-x. 2 root root 6 May  2 06:36 AppStream
drwxr-xr-x. 2 root root 6 May  2 06:36 BaseOS
[ronaldo@lamp-ansible ~]$
```

Copy Packages

cp -r /mnt/iso/BaseOS/* /localrepo/BaseOS/

cp -r /mnt/iso/AppStream/* /localrepo/AppStream/

The above two commands to be ignored for this reason:

```
Desktop          dh2 dh4 Documents Music      Public  Videos
[ronaldo@lamp-ansible ~]$ sudo cp -r /mnt/iso/BaseOS/* /localrepo/BaseOS/
[ronaldo@lamp-ansible ~]$ sudo cp -r /mnt/iso/AppStream/* /localrepo/AppStream/
cp: error writing '/localrepo/AppStream/Packages/golang-bin-1.26.2-1.el9.x86_64.rpm': No space left on device
cp: cannot create regular file '/localrepo/AppStream/Packages/golang-docs-1.26.2-1.el9.noarch.rpm': No space left on device
cp: cannot create regular file '/localrepo/AppStream/Packages/golang-misc-1.26.2-1.el9.noarch.rpm': No space left on device
cp: cannot create regular file '/localrepo/AppStream/Packages/golang-race-1.26.2-1.el9.x86_64.rpm': No space left on device
cp: cannot create regular file '/localrepo/AppStream/Packages/golang-src-1.26.2-1.el9.noarch.rpm': No space left on device
cp: cannot create regular file '/localrepo/AppStream/Packages/golang-tests-1.26.2-1.el9.noarch.rpm': No space left on device
cp: cannot create regular file '/localrepo/AppStream/Packages/gom-0.4-6.el9.i686.rpm': No space left on device
cp: cannot create regular file '/localrepo/AppStream/Packages/gom-0.4-6.el9.x86_64.rpm': No space left on device
cp: cannot create regular file '/localrepo/AppStream/Packages/google-carlito-fonts-1.103-0.18.20130920.el9.noarch.rpm': No space left on device
cp: cannot create regular file '/localrepo/AppStream/Packages/google-crosextra-caladea-fonts-1.002-0.15.20130214.el9.noarch.rpm': No space left on device
cp: cannot create regular file '/localrepo/AppStream/Packages/google-droid-sans-fonts-20200215-11.el9.2.noarch.rpm': No space left on device
cp: cannot create regular file '/localrepo/AppStream/Packages/google-droid-sans-mono-fonts-20200215-11.el9.2.noarch.rpm': No space left on device
cp: cannot create regular file '/localrepo/AppStream/Packages/google-droid-serif-fonts-20200215-11.el9.2.noarch.rpm': No space left on device
```

Explanation of my report

During the process of copying repository data from the mounted ISO to a local directory, an error occurred indicating “No space left on device.” This was due to insufficient disk space, as the BaseOS and AppStream repositories require several gigabytes of storage. The issue was resolved by opting to use the mounted ISO directly as a repository source, eliminating the need for large data duplication.

Configure Repo File directly(no data duplication anymore)

sudo nano /etc/yum.repos.d/baseos.repo

[BaseOS]

baseurl=file:///localrepo/BaseOS

enabled=1

gpgcheck=0

sudo nano /etc/yum.repos.d/appstream.repo

[AppStream]

baseurl=file:///localrepo/AppStream

enabled=1

gpgcheck=0

```
[ronaldo@lamp-ansible ~]$  
[ronaldo@lamp-ansible ~]$ sudo nano /etc/yum.repos.d/baseos.repo  
[sudo] password for ronaldo:  
[ronaldo@lamp-ansible ~]$ sudo nano /etc/yum.repos.d/appstream.repo  
[sudo] password for ronaldo:  
[ronaldo@lamp-ansible ~]$ sudo dnf clean all  
21 files removed  
[ronaldo@lamp-ansible ~]$ df -h  
Filesystem      Size  Used Avail Use% Mounted on  
devtmpfs        821M   0 821M   0% /dev  
tmpfs           854M   0 854M   0% /dev/shm  
tmpfs           342M  11M 332M   4% /run  
/dev/mapper/cs-root 27G  22G  5.7G  79% /  
/dev/sdal       960M  519M  442M  54% /boot  
tmpfs           171M  36K 171M   1% /run/user/1000  
/dev/loop0      15G   15G   0 100% /mnt/iso  
[ronaldo@lamp-ansible ~]$ sudo dnf makecache  
CentOS Stream 9 AppStream                6.6 MB/s | 8.4 MB    00:01  
CentOS Stream 9 BaseOS                   6.1 MB/s | 2.2 MB    00:00  
CentOS Stream 9 - BaseOS                  2.0 MB/s | 8.9 MB    00:04  
CentOS Stream 9 - AppStream                2.1 MB/s | 27 MB     00:13  
CentOS Stream 9 - Extras packages          9.8 kB/s | 21 kB     00:02  
Last metadata expiration check: 0:00:01 ago on Sat 02 May 2026 10:29:35 AM SAST.  
Metadata cache created.  
[ronaldo@lamp-ansible ~]$ sudo dnf repolist  
repo id                                repo name  
AppStream                              CentOS Stream 9 AppStream  
BaseOS                                  CentOS Stream 9 BaseOS  
appstream                              CentOS Stream 9 - AppStream  
baseos                                  CentOS Stream 9 - BaseOS  
extras-common                          CentOS Stream 9 - Extras packages  
[ronaldo@lamp-ansible ~]$ sudo install httpd -y
```

Clean old cache:

sudo dnf clean all

Rebuild Repository metadata

sudo dnf makecache.

Verify repository setup

sudo dnf repolist

Expected output : baseOS ,Appstream etc.

Test installation:

sudo dnf httpd -y

```
Try 'install --help' for more information.
[ronaldo@lamp-ansible ~]$ sudo dnf install httpd -y
Last metadata expiration check: 0:03:02 ago on Sat 02 May 2026 10:29:35 AM SAST.
Dependencies resolved.
=====
Package                                Architecture      Version           Repository        Size
=====
Installing:
httpd                                   x86_64            2.4.62-13.el9     AppStream         47 k
Installing dependencies:
apr                                     x86_64            1.7.0-12.el9      AppStream         123 k
apr-util                               x86_64            1.6.1-23.el9      AppStream         95 k
apr-util-bdb                           x86_64            1.6.1-23.el9      AppStream         13 k
centos-logos-httpd                     noarch            90.9-1.el9        AppStream         1.5 M
httpd-core                             x86_64            2.4.62-13.el9     AppStream         1.5 M
httpd-filesystem                       noarch            2.4.62-13.el9     AppStream         12 k
httpd-tools                             x86_64            2.4.62-13.el9     AppStream         81 k
Installing weak dependencies:
apr-util-openssl                       x86_64            1.6.1-23.el9      AppStream         15 k
mod_http2                              x86_64            2.0.26-5.el9      AppStream        163 k
mod_lua                                x86_64            2.4.62-13.el9     AppStream         59 k
Transaction Summary
=====
Install 11 Packages

Total size: 3.6 M
Installed size: 8.6 M
Downloading Packages:
```

Conclusion. An offline repo had been finally configured in my CentOS9 stream lamp-ansible server.

```
Verifying      : httpd-2.4.62-13.el9.x86_64                                6/11
Verifying      : httpd-core-2.4.62-13.el9.x86_64                          7/11
Verifying      : httpd-filesystem-2.4.62-13.el9.noarch                     8/11
Verifying      : httpd-tools-2.4.62-13.el9.x86_64                        9/11
Verifying      : mod_http2-2.0.26-5.el9.x86_64                           10/11
Verifying      : mod_lua-2.4.62-13.el9.x86_64                             11/11

Installed:
  apr-1.7.0-12.el9.x86_64          apr-util-1.6.1-23.el9.x86_64          apr-util-bdb-1.6.1-23.el9.x86_64
  apr-util-openssl-1.6.1-23.el9.x86_64  centos-logos-httpd-90.9-1.el9.noarch  httpd-2.4.62-13.el9.x86_64
  httpd-core-2.4.62-13.el9.x86_64      httpd-filesystem-2.4.62-13.el9.noarch  httpd-tools-2.4.62-13.el9.x86_64
  mod_http2-2.0.26-5.el9.x86_64        mod_lua-2.4.62-13.el9.x86_64

Complete!
[ronaldo@lamp-ansible ~]$ sudo /home/ronaldo/CentOS-Stream-9-latest-x86_64-dvd1.iso /mnt/iso iso9660 loop,ro 0 0
[sudo] password for ronaldo:
sudo: /home/ronaldo/CentOS-Stream-9-latest-x86_64-dvd1.iso: command not found
[ronaldo@lamp-ansible ~]$ /home/ronaldo/CentOS-Stream-9-latest-x86_64-dvd1.iso /mnt/iso iso9660 loop,ro 0 0
-bash: /home/ronaldo/CentOS-Stream-9-latest-x86_64-dvd1.iso: Permission denied
[ronaldo@lamp-ansible ~]$ sudo /home/ronaldo/CentOS-Stream-9-latest-x86_64-dvd1.iso /mnt/iso iso9660 loop,ro 0 0
sudo: /home/ronaldo/CentOS-Stream-9-latest-x86_64-dvd1.iso: command not found
[ronaldo@lamp-ansible ~]$ sudo mount -a
[ronaldo@lamp-ansible ~]$ ls /mnt/iso
AppStream BaseOS EFI EULA extra_files.json images isolinux LICENSE media.repo
[ronaldo@lamp-ansible ~]$ sudo dnf repolist
repo id                                repo name
AppStream                              CentOS Stream 9 AppStream
BaseOS                                 CentOS Stream 9 BaseOS
appstream                             CentOS Stream 9 - AppStream
baseos                                CentOS Stream 9 - BaseOS
extras-common                          CentOS Stream 9 - Extras packages
[ronaldo@lamp-ansible ~]$ sudo dnf config-manager --set-disabled \*
[ronaldo@lamp-ansible ~]$
```

To setup a repo Server via HTTP for multi-server environment.

Idea: An HTTP-based repository server configured with Apache HTTP Server provides centralized package hosting for client systems running CentOS Stream 9. It enables local access to RPM packages via standard package managers (e.g., dnf), reducing latency and external network dependency. This setup ensures version consistency, supports controlled update rollouts, and is suitable for both online and air-gapped environments.

Now: All systems install packages locally No external dependency

dnf install httpd -y

```
baseOS
[ronaldo@lamp-ansible ~]$ sudo dnf install httpd -y
Last metadata expiration check: 0:01:20 ago on Sat 02 May 2026 12:56:05 PM SAST.
Package httpd-2.4.62-13.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[ronaldo@lamp-ansible ~]$
```

systemctl enable --now httpd

Systemctl status httpd

```
baseOS
[ronaldo@lamp-ansible ~]$ sudo dnf install httpd -y
Last metadata expiration check: 0:01:20 ago on Sat 02 May 2026 12:56:05 PM SAST.
Package httpd-2.4.62-13.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[ronaldo@lamp-ansible ~]$ sudo systemctl enable --now httpd
[sudo] password for ronaldo:
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[ronaldo@lamp-ansible ~]$ systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Sat 2026-05-02 14:15:50 SAST; 1min 12s ago
     Docs: man:httpd.service(8)
  Main PID: 60165 (httpd)
    Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
     Tasks: 177 (limit: 10501)
    Memory: 14.2M (peak: 14.6M)
       CPU: 220ms
    CGroup: /system.slice/httpd.service
            └─60165 /usr/sbin/httpd -DFOREGROUND
              └─60167 /usr/sbin/httpd -DFOREGROUND
                └─60168 /usr/sbin/httpd -DFOREGROUND
                  └─60169 /usr/sbin/httpd -DFOREGROUND
                    └─60170 /usr/sbin/httpd -DFOREGROUND

May 02 14:15:50 lamp-ansible systemd[1]: Starting The Apache HTTP Server...
May 02 14:15:50 lamp-ansible httpd[60165]: Server configured, listening on: port 80
May 02 14:15:50 lamp-ansible systemd[1]: Started The Apache HTTP Server.
[ronaldo@lamp-ansible ~]$
```

sudo ln -s /mnt/iso /var/www/html/repo

This command creates a **symbolic link (soft link)** between the mounted ISO directory and the Apache web server's document root.

Breakdown

sudo

→ Executes the command with administrative privileges.

ln -s

→ Creates a symbolic (soft) link, not a physical copy of files.

/mnt/iso

→ The source directory where the CentOS ISO is mounted.

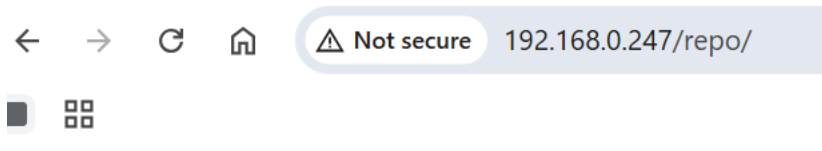
/var/www/html/repo

→ The destination path served by the Apache web server.

```
[sudo] password for ronaldo:
[ronaldo@lamp-ansible ~]$ ls -l /var/www/html
total 0
[ronaldo@lamp-ansible ~]$ sudo ln -s /mnt/iso /var/www/html/repo
[ronaldo@lamp-ansible ~]$ ls -l /var/www/html
total 0
lrwxrwxrwx. 1 root root 8 May  2 15:54 repo -> /mnt/iso
[ronaldo@lamp-ansible ~]$
```

Repo becomes accessible over network:

<http://server-ip/repo>



Index of /repo

	Name	Last modified	Size	Description
	Parent Directory		-	
	AppStream/	2026-04-27 06:13	-	
	BaseOS/	2026-04-27 06:13	-	
	EFI/	2026-04-27 06:02	-	
	EULA	2026-04-27 06:13	299	
	LICENSE	2026-04-07 23:09	18K	
	extra_files.json	2026-04-27 06:13	745	
	images/	2026-04-27 06:02	-	
	isolinux/	2026-04-27 06:02	-	
	media.repo	2026-04-27 06:13	88	